

2

Attendance checker

```
import java.util.Scanner;
```

Public class Main {

```
public static void main (String[] args) {
```

```
Scanner sc = new Scanner(System.in);
```

system.out.println("Enter total class conducted");

```
int totalClasses = sc.nextInt();
```

```
system.out.print("Enter classes attended: ");
```

~~return~~ int attended = sc.nextInt();
$$\text{double percentage} = (\text{double}) \text{attended} / \text{total classes} \times 100;$$

system.out.println("Attendance percentage:");

~~is percentag~~ + percentage + "%";

if (percentage ≥ 75) {

system.out.println("Eligible for Exam");

3 eise f

```
system.out.println("not Eligible");
```

3 / output

Enter total classes conducted.	17
--------------------------------	----

Enter classes attended: 5

Attendance percentage: 41.57%

Enter classes attended: 5

Attendance percentage: 41.57%

Not Eligible

Teacher's Signature.....

2)

AI Age predictor

3)

```
import java.util.Scanner;
import java.time.Year;
```

```
public class Main {
    public static void main(String[] Args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your current age");
        int age = sc.nextInt();

        System.out.println("Age after 10 years: " + (age + 10));
        System.out.println("Age after 25 years: " + (age + 25));
        System.out.println("Age after 50 years: " + (age + 50));

        int currentYear = Year.now().getValue();
        int yearTurn100 = currentYear + (100 - age);

        System.out.println("You will turn 100 in the year: "
            + yearTurn100);
    }
}
```

Output

```
? Enter your current age: 22
Age after 10 years: 32
Age after 25 years: 47
Age after 50 years: 72
You will turn 100 in year: 2104
```

3) Fitness challenge Tracker

```
import java.util.Scanner;
public class main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        int total = 0;
        int highest = 0;
        System.out.println ("Enter steps for 7 days:");
        for (int i = 1; i <= 7; i++) {
            System.out.print ("Day " + i + ": ");
            int steps = sc.nextInt();
            total += steps;
            if (steps > highest) {
                highest = steps;
            }
        }
        double average = total / 7.0;
        System.out.println ("Total steps = " + total);
        System.out.println ("Average steps = " + average);
        System.out.println ("Highest steps = " + highest);
    }
}
```

~~end~~

Output

Enter steps for 7 days:

Day 1: 3333

Day 2: 4444

Day 3: 5555

Day 4: 6666

Day 5: 7777

Day 6: 8888

Day 7: 9999

Total steps = 46662

Average steps = 6666.0

Highest steps = 9999

4) prime Number check

```

import java.util.Scanner
public class Main {
    public static void main (String[] args);
    Scanner sc = new Scanner (System.in);
    System.out.print ("Enter a number:");
    int num = sc.nextInt ();
    boolean isprime = true;
    if (num <= 1) {
        isprime = false;
    } else {
        for (int i = 2; i <= num/2; i++) {
            if (num % i == 0) {
                isprime = false;
                break;
            }
        }
        if (isprime) {
            System.out.println (num + " is prime!");
        } else {
            System.out.println (num + " is NOT prime!");
        }
    }
}

```

output

Enter a number: 11

11 is prime.

Teacher's Signature

5) Fibonacci series

```
import java.util.Scanner;
public class main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter n: ");
        int n = sc.nextInt();
        int a = 0, b = 1;
        System.out.print("Fibonacci series: ");
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
    }
}
```

3
3
3
3

Output

Enter n: 9

Fibonacci series: 0 1 1 2 3 5 8 13 21

Logical operators check

```
import java.util.Scanner;
public class main {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number:");
        int a = sc.nextInt();
        System.out.print("Enter second number:");
        int b = sc.nextInt();
        System.out.print("Enter third number:");
        int c = sc.nextInt();
        boolean result = (c == (a+b));
        System.out.println("Is third number the
            sum of first two?" + result);
    }
}
```

output

```
Enter first number: 4
Enter second number: 7
Enter third number: 9
Is third number the sum of first
two? false
```

Multiplication Table

```
import java.util.Scanner;
public class main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print("Enter a number:");
        int num = sc.nextInt();
        for (int i = 1; i <= 10; i++) {
            System.out.println (num + " x " + i + " = "
                                + (num * i));
        }
    }
}
```

Output

Enter a number: 10

$$10 \times 1 = 10$$

$$10 \times 2 = 20$$

$$10 \times 3 = 30$$

$$10 \times 4 = 40$$

$$10 \times 5 = 50$$

$$10 \times 6 = 60$$

$$10 \times 7 = 70$$

$$10 \times 8 = 80$$

$$10 \times 9 = 90$$

$$10 \times 10 = 100$$

8) Right-Angled Triangle pattern

```

import java.util.Scanner;
public class main {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of rows:");
        int rows = sc.nextInt();
        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}

```

Output

Enter number of rows:

*

**

9)

Pyramid pattern

```

import java.util.Scanner
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter number of rows:");
        int rows = sc.nextInt ();
        for (int i = 1; i <= rows; i++) {
            for (int j = 1; j <= rows - i; j++) {
                System.out.print (" ");
            }
            for (int k = 1; k <= (2 * i - 1); k++) {
                System.out.print ("*");
            }
            System.out.println ();
        }
    }
}

```

output

Enter number of rows: 5

*

* * *

* * * * *

* * * * * * * *

* * * * * * * * *

Teacher's Signature.....

palindrome number

```

import java.util.Scanner;

public class Main {
    public static boolean void main isPalindrome (int x) {
        int original = x;
        int reverse = 0;
        while (x > 0) {
            int digit = x % 10;
            reverse = reverse * 10 + digit;
            x /= 10;
        }
        return original == reverse;
    }

    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number:");
        int num = sc.nextInt();
        System.out.println(isPalindrome(num));
    }
}

```

output

Enter a number: 7
true

11)

Basic calculator

```
import java.util.Scanner;
public class Main {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print ("Enter first number:");
        double a = sc.nextDouble();
        System.out.print ("Enter second number:");
        double b = sc.nextDouble();
        System.out.print ("Enter operator (+, -, *, /)");
        char op = sc.next().charAt(0);
        switch (op) {
            case '+':
                System.out.println ("Result = " + (a+b));
                break;
            case '-':
                System.out.println ("Result = " + (a-b));
                break;
            case '*':
                System.out.println ("Result = " + (a*b));
                break;
            case '/':
                if (b != 0)
                    System.out.println ("Result = " + (a/b));
```

else

system.out.println("Division by zero is not allowed");

break;

default;

system.out.println("Invalid operator");

}

Output

Enter first number: 4

Enter second number: 5

Enter operator (+, -, *, /): *

Result = 20.00

12)

Urade calculator

```

import java.util.Scanner
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter marks: ");
        int marks = sc.nextInt();
        String grade =
            (marks >= 90) ? "A+" :
            (marks >= 80) ? "A" :
            (marks >= 70) ? "B" :
            (marks >= 60) ? "B" :
            (marks >= 50) ? "C+" :
            (marks >= 40) ? "C" :
            "F";
        System.out.println("Urade = " + grade);
    }
}

```

Output

Enter marks: 33

Urade = F